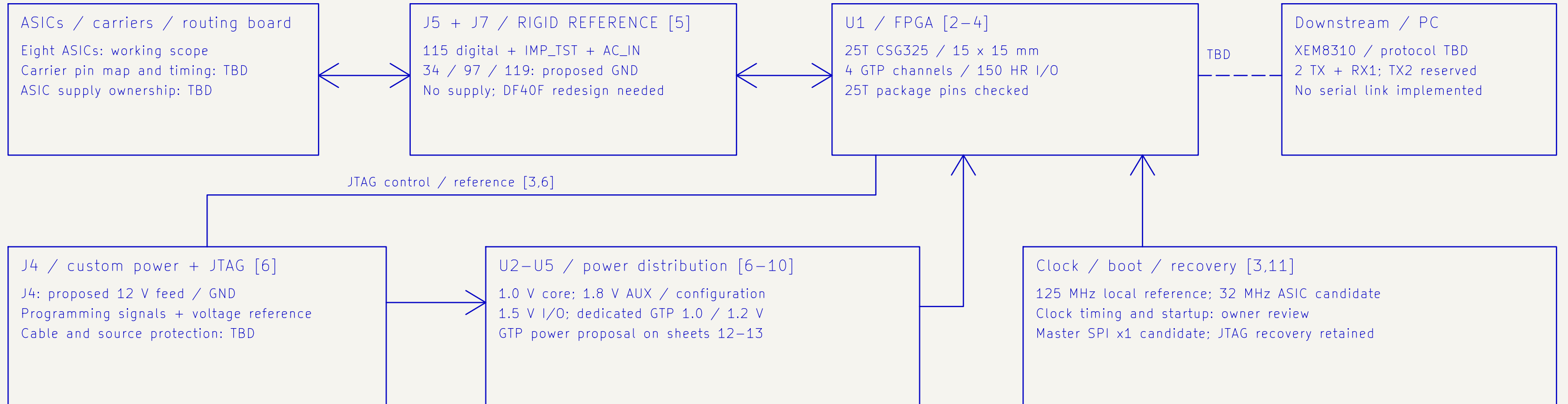


FPGA BOARD / OVERALL CONNECTIONS

29 September 2026 | XC7A25T-2CSG325I | Eight copper layers | Unrouted review

Data / returned clocks enter FPGA; generated clocks / SPI / reset / control return toward ASICs. AC_IN has no FPGA path.



SIGNAL PRIORITIES

- 1 Capture data + returned clocks: derive per-ASIC setup/hold/skew groups. Include carrier and connector delay.
- 2 GTP and reference pairs, if selected: impedance, intra-pair skew, loss, reference continuity and rail filtering.
- 3 SPI / resets / record control: mode, pulse widths, safe defaults and startup order. Slow does not mean optional.
- 4 Power and ground: load/return paths, sequencing, decoupler loop area, connector budget and cable transients.

SCHEMATIC NAVIGATION – one overview plus thirteen direct detail sheets

02 / 25T I/O banks 14 / 15 / 34	03 / Master SPI x1 mode and clock hold
04 / 25T core / auxiliary / ground	05 / two DF40T-60 ASIC connectors
06 / Actual J4 input + core/AUX power	07 / I/O power; U4 now on sheet 13
08 / Core / BRAM decoupling review	09 / Auxiliary / configuration decoupling
10 / ASIC bank decoupling review	11 / 32 Mbit flash boot candidate
12 / GTP pins + manual INIT_B hold	13 / Dedicated GTP 1.0 V + 1.2 V power
14 / GTP AC coupling and 125 MHz reference	

Overview lines explain functional paths. Detailed electrical connectivity uses identical global net names across sheets.

8 Cu layers; rigid connector reference; no routed copper. 36 x 38 mm; floating DF40F and power entry require redesign.

UNROUTED POWER PROPOSAL – NOT QUALIFIED

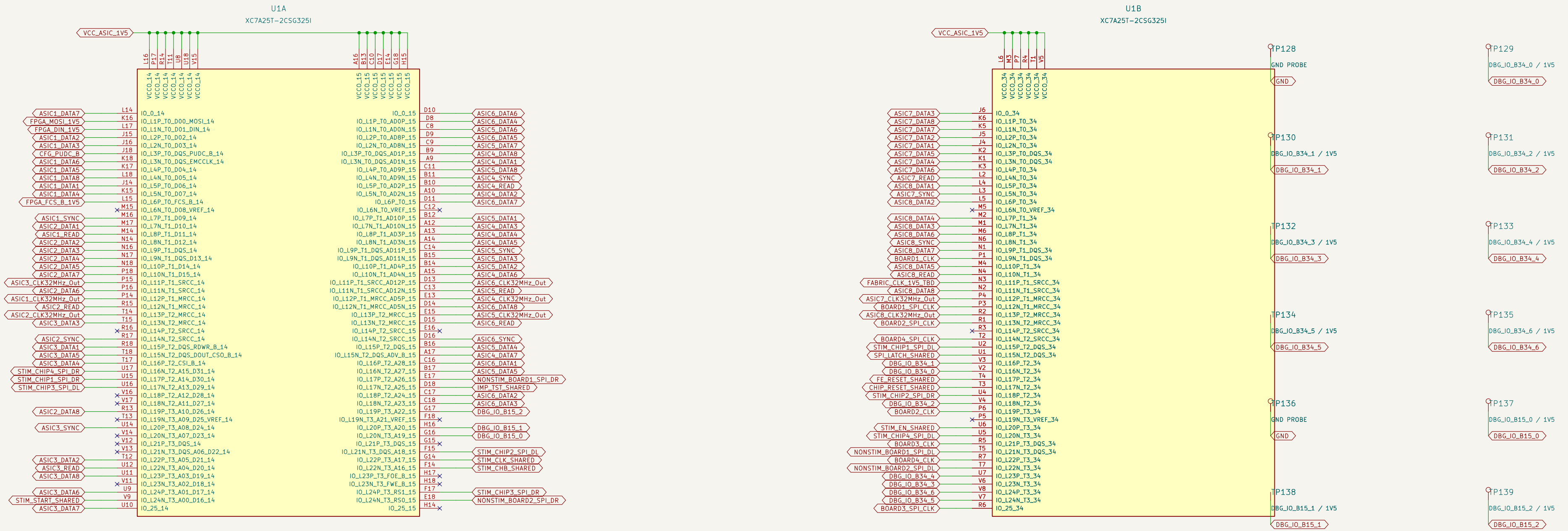
Sheet: /
File: FPGA25T_8L_Unrouted.kicad_sch

Title: FPGA board – overall signal and power connections

Size: A3	Date: 2026-09-29	Rev: 25T R1
KiCad E.D.A. 10.0.6		Id: 1/14

02 / FPGA digital I/O – what each signal group does

Three 1.5 V I/O banks; proposed ball assignments. Pin names connect to the same names on J5, sheet 5.



ASIC DATA + RETURNED CLOCKS

Each ASIC sends digital data and a capture clock to U1.

Use that ASIC clock to define its data timing group.

Match delays within the required group after Gerald confirms setup/hold timing, including carrier delays.

PROPOSED BANK ALLOCATION

ASIC 1–3: bank 14. ASIC 4–6: bank 15. ASIC 7–8: bank 34.

All three use the same 1.5 V supply in this draft.

Vivado pin/clock checks and carrier pin-map review remain.

COMMANDS SENT BACK TO THE ASICs

SPI writes configuration; reset and recording controls set the operating state. Generated clocks go outward. Confirm direction, polarity and startup defaults before routing. Low-speed controls still need valid timing.

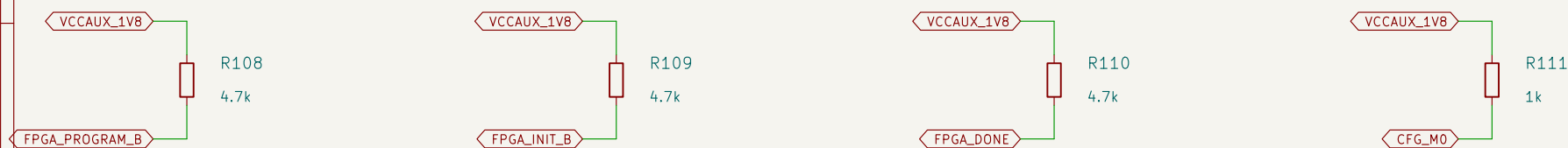
CLOCK SOURCE + UNCONNECTED PINS

FABRIC_CLK_1V5_TBD has no oscillator connected.
The 32 MHz ASIC clock source remains to be selected.
NC means intentionally unconnected in this proposal;
it does not mean that a carrier signal can be omitted.

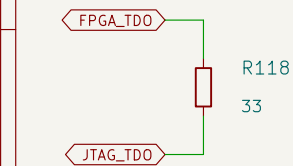
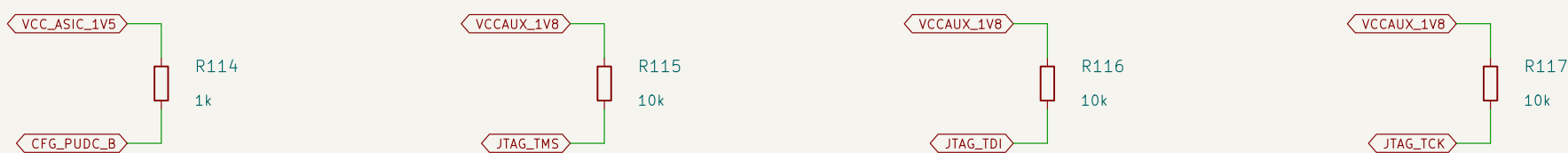
03 / Programming and startup defaults

Master SPI x1 candidate: M[2:0] = 001; bank 0 = 1.8 V; CFGBVS = GND. JTAG remains available.

R108–R111 / configuration pull-ups

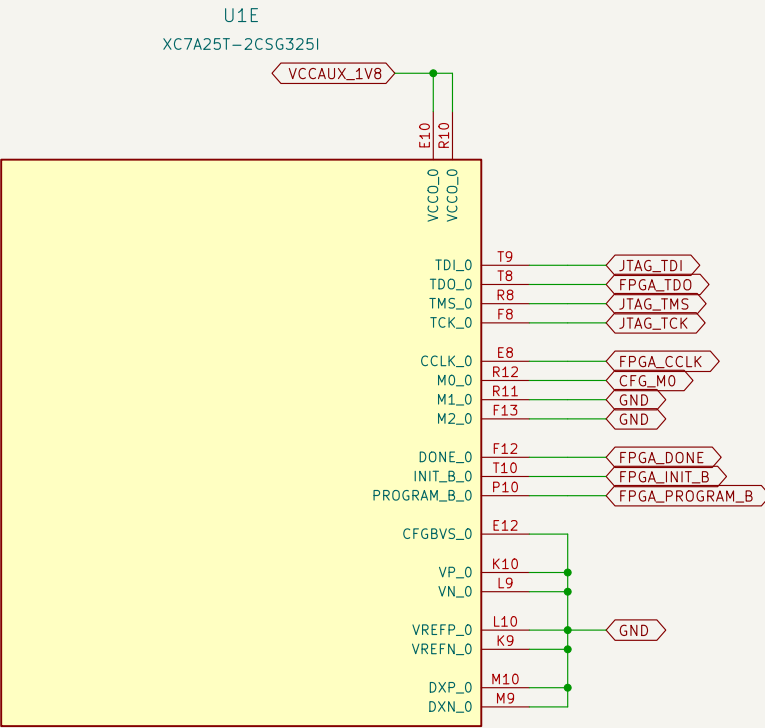


R114–R117 / defined startup states

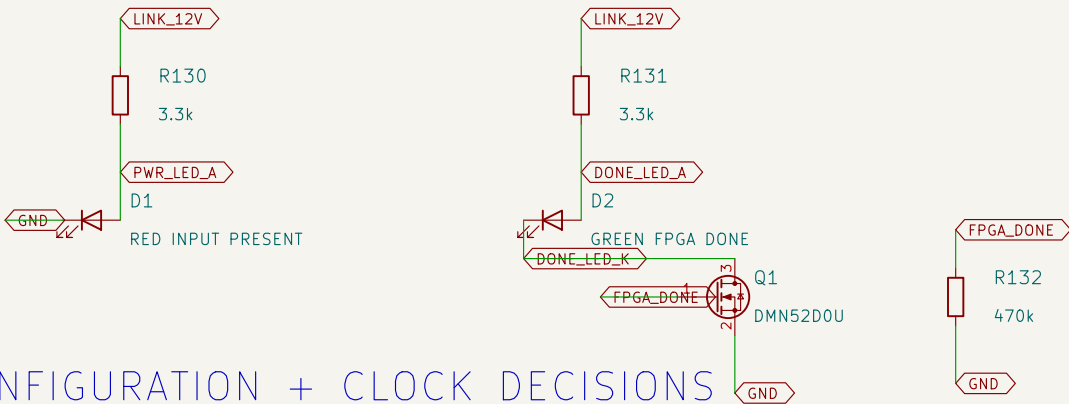


R118 / SERIES RESISTOR

33 ohm in the JTAG TDO return path.
It is a damping provision; final placement and value depend on the cable and edge-rate review.



32 MHz ASIC clock source and fabric clock plan remain TBD

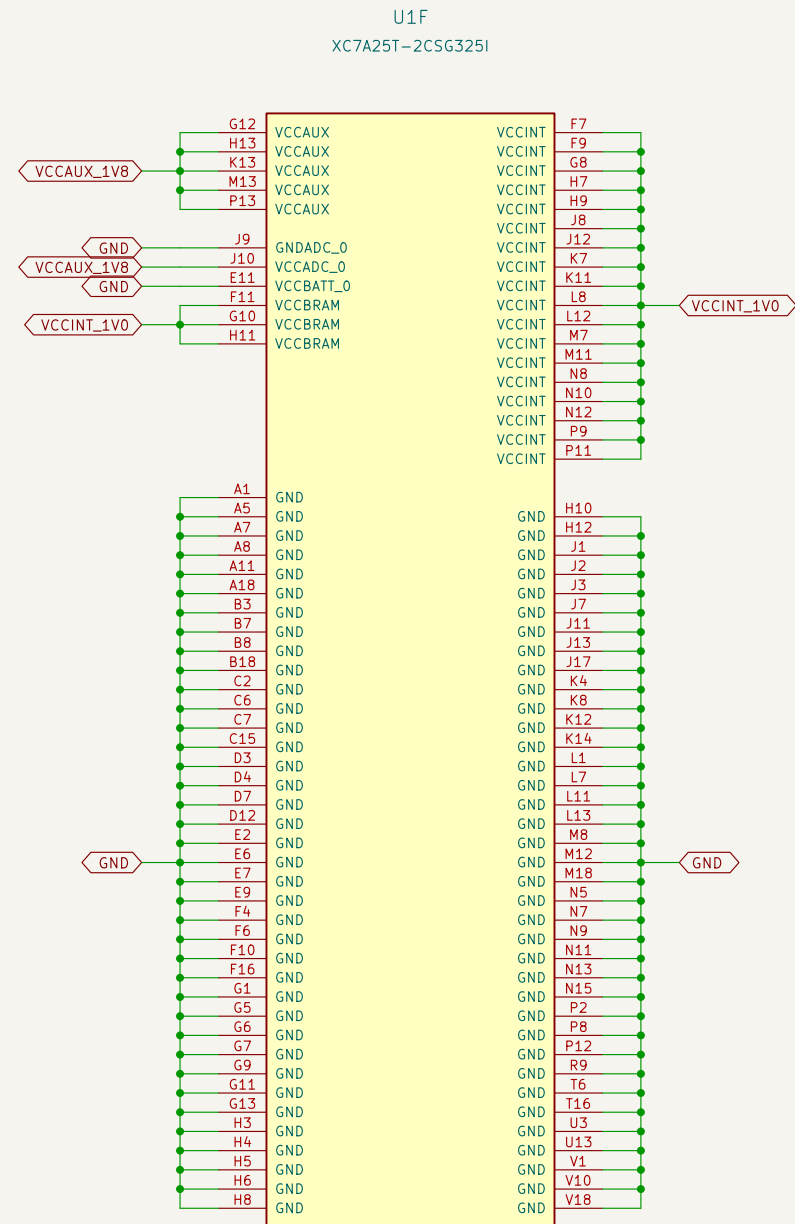


U1 CONFIGURATION + CLOCK DECISIONS

JTAG remains an alternative programming path to flash.
J4 pin 1 reports the board 1.8 V logic reference;
it is not an external power input. See sheet 6.
Choose a fabric clock and a dedicated GTP reference.
100 MHz remains a candidate, not a finalized part.

04 / FPGA power pins and ground

Same net name means an electrical connection across sheets; power entry and regulators are on sheets 6–7.



1.0 V / CORE + BLOCK MEMORY

U2 supplies the FPGA logic and internal memory.

R9 is in the path to the distributed core rail.

Core capacitor groups are shown on sheet 8.

1.8 V / AUXILIARY + CONFIGURATION

U3 feeds auxiliary functions, XADC and bank 0.

R122 is in the distributed auxiliary supply path.

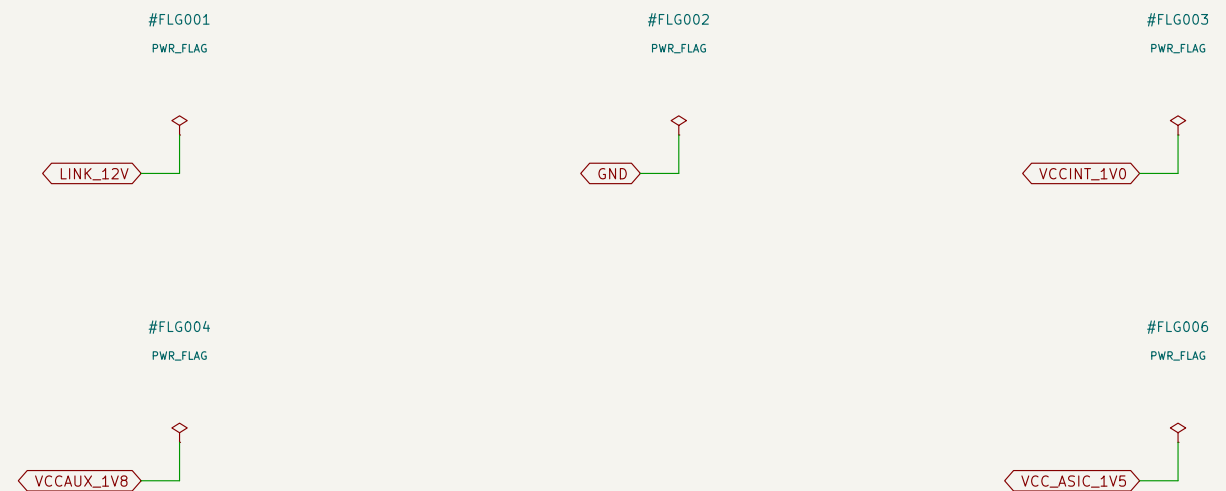
Bank 0 also sets the JTAG reference voltage.

OTHER DOMAINS + GROUND

1.5 V I/O bank pins are on sheet 2.

GTP analog power pins are on sheet 12.

Ground is the common current–return connection.



PWR_FLAG tells the schematic checker a rail is driven.

It is not a regulator or a proof of safe power-up.

UNROUTED POWER PROPOSAL – NOT QUALIFIED

Sheet: /04 / 25T core / auxiliary / ground/
File: review_04_fpga_power.kicad_sch

Title: 04 / 25T core, auxiliary, ground and source assertions

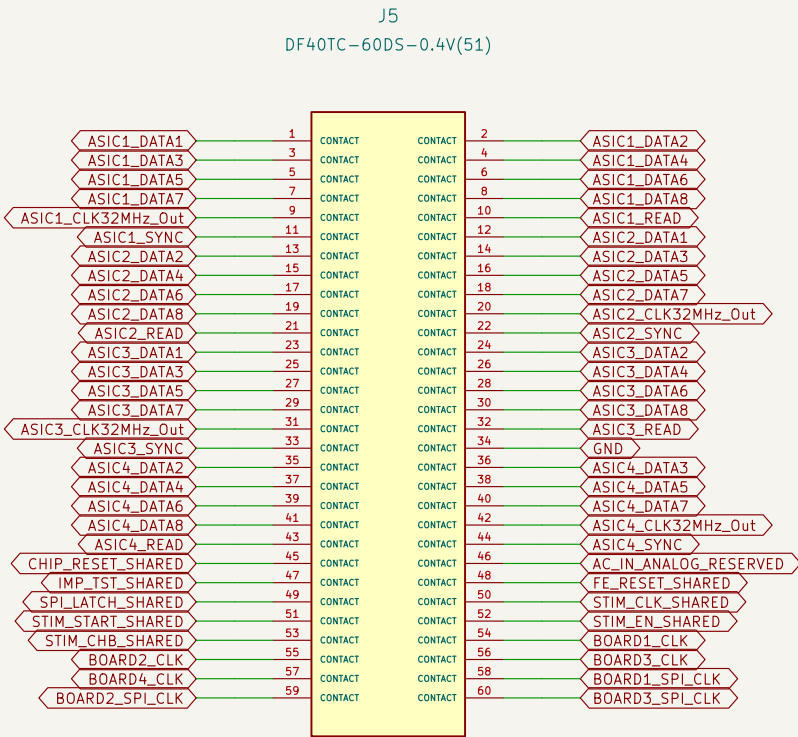
Size: A3	Date: 2026-09-29	Rev: R5 GTP power
KiCad E.D.A. 10.0.6		Id: 4/14

05 / Two physical DF40T 60–contact ASIC connectors

116 digital signals + 1 analog reservation + 3 proposed grounds. This is a proposed contact map.

J5 / CONTACTS 1–60

First physical DF40T connector; original contacts 1–60.



SIGNALS + RETURN PATHS

Names match the FPGA nets on sheet 2.

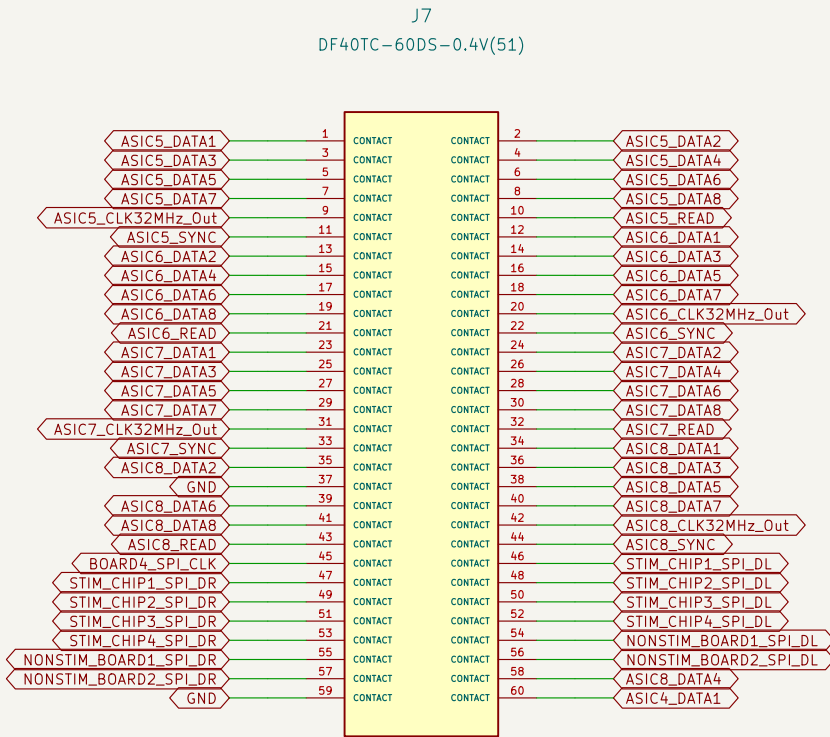
J5.34 / J7.37 / J7.59 are proposed ground contacts.

Their return–path adequacy still needs review.

AC_IN is reserved; it has no FPGA input path.

J7 / CONTACTS 1–60

Second physical DF40T connector; old contacts 61–120 are now J7.1–60.



POWER + MECHANICAL INTERFACE

No contact is assigned to power the ASICs.

Confirm carrier supplies and pin–1 orientation.

Proposed mate on routing board: DF40TC–60DP–0.4V(51) (two pieces).

Two rigid–board pairs conflict with Hirose ETAD–H1015; mechanical solution pending.

UNROUTED POWER PROPOSAL – NOT QUALIFIED

Sheet: /05 / two DF40T–60 ASIC connectors/
File: review_05_df40.kicad_sch

Title: Two DF40T 60–contact ASIC connectors – review

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026–09–29

Rev: R6 connector split
Id: 5/14

06 / Custom power input, core and auxiliary supplies

Read left to right: J4 brings 12 V in; each buck regulator makes a lower–voltage rail for the FPGA.

EXTERNAL HARDWARE

Regulated 12 V supply + 1.8 V JTAG probe.

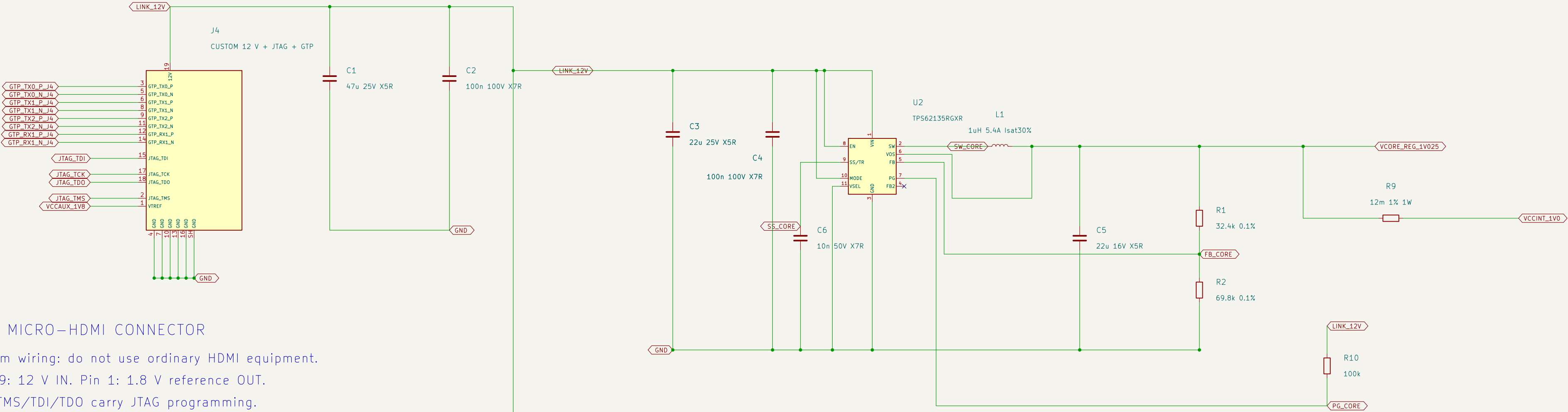
A custom breakout combines them into J4.

The breakout and cable are not yet designed.

U2 / CORE + MEMORY: ~1.0249 V BEFORE R9

L1/C5 smooth the switched output; R1/R2 set its voltage.

C3/C4 bypass VIN. C6 controls the soft–start ramp.



J4 / MICRO–HDMI CONNECTOR

Custom wiring: do not use ordinary HDMI equipment.

Pin 19: 12 V IN. Pin 1: 1.8 V reference OUT.

TCK/TMS/TDI/TDO carry JTAG programming.

Three TX pairs and one RX pair use 100 nF AC caps on sheet 14.

Supply negative + probe ground join board GND.

C1/C2 / INPUT BYPASS

Capacitors provide local charge at the input

and help reduce fast voltage disturbance.

Fuse, polarity and surge protection are absent.

Cable current and inrush need qualification.

PG / STARTUP ENABLE CHAIN

U2 starts first; PG_CORE then enables U3.

PG_AUX enables U5 and new GTP regulator U7.

R10/R11 pull these PG signals toward 12 V;

do not connect them directly to FPGA I/O.

R9/R122 / SERIES SUPPLY PATH

Feedback senses BEFORE these resistors.

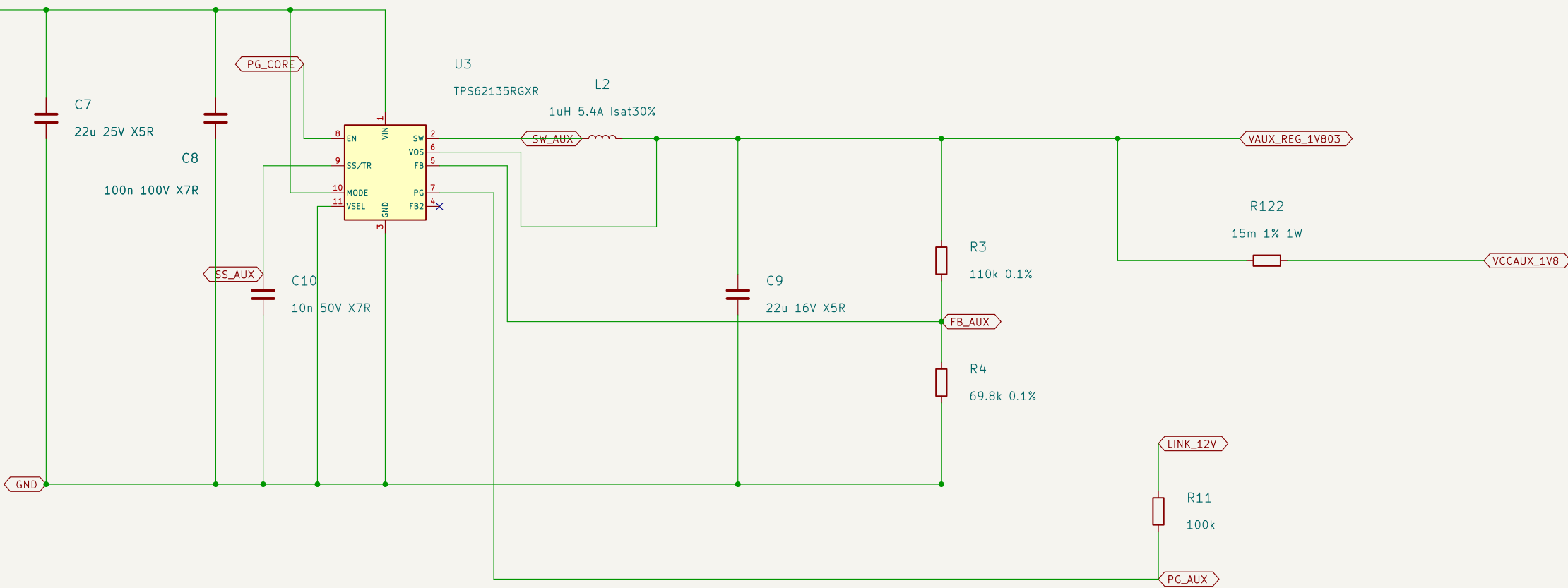
Their voltage drop grows with load current.

Verify voltage at the FPGA, not only at U2/U3.

U3 / AUXILIARY + BANK 0: ~1.8032 V BEFORE R122

L2/C9 smooth the output; R3/R4 set its voltage.

C7/C8 bypass VIN. C10 controls the soft–start ramp.



VOS/FB sense the local output. Distributed bypass: sheets 8–9. GTP supply proposal: sheet 13.

UNROUTED POWER PROPOSAL – NOT QUALIFIED

Sheet: /06 / Actual J4 input + core/AUX power/
File: review_06_core_aux_regulators.kicad_sch

Title: 06 / Actual power entry, core and auxiliary circuits

Size: A2

Date: 2026–09–29

Rev: R5 GTP power

KiCad E.D.A. 10.0.6

Id: 6/14

07 / FPGA I/O supply – U4 repurposed on sheet 13

U5 receives the common input bus. Its output powers FPGA I/O only.

1.5 V / FPGA I/O BANK POWER

U5 sets the electrical supply for banks 14, 15 and 34, which face the ASICs.

This rail does not power the ASIC chips.

J5 has no assigned ASIC supply contact.

U5 / ~1.5083 V / I/O BANKS 14, 15 AND 34

L4/C17: output filter. R7/R8: voltage feedback.

C15/C16: input bypass. C18: soft–start capacitor.

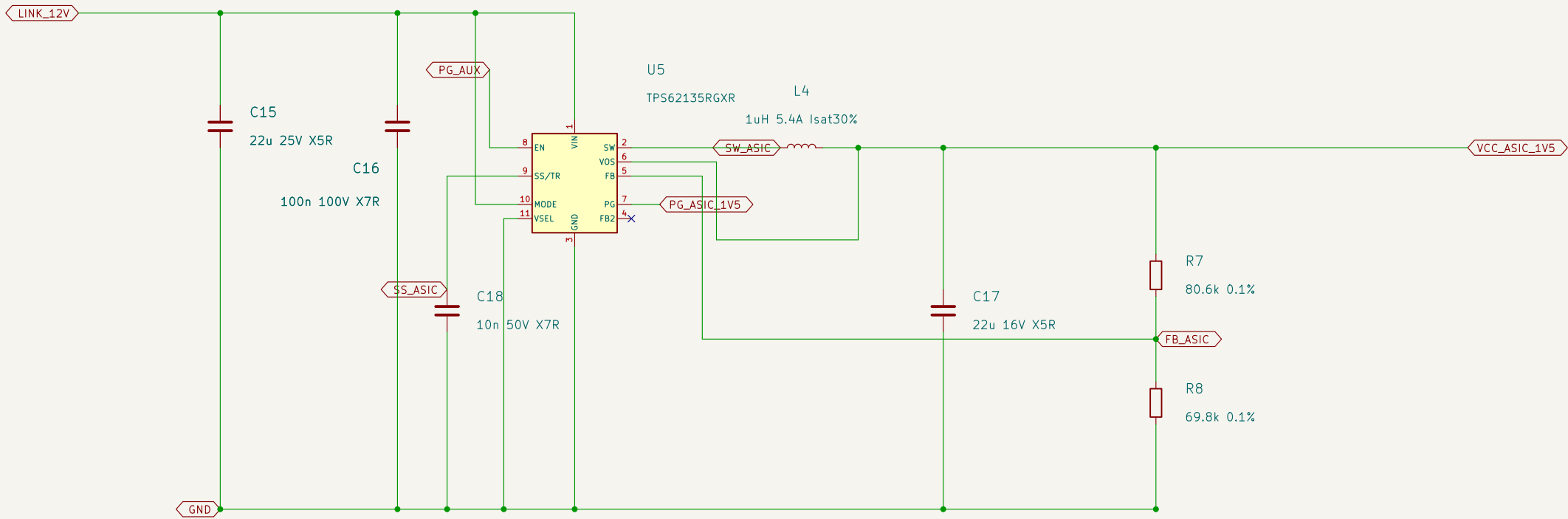
HOW A BUCK SECTION WORKS

The IC switches energy into its inductor.

The inductor and output capacitor smooth it.

The feedback divider tells the IC the voltage.

Input bypass supplies short current pulses.



EN / PG / SOFT START

EN starts the regulator when PG_AUX rises.

The soft–start capacitor controls the ramp.

U5 PG remains unconnected; U4 PG is on sheet 13.

U4 HAS MOVED TO THE GTP POWER CIRCUIT

The unused 2.5 V rail and C57–C63 have been removed.

U4 now supplies GTP 1.2 V through its own filter on sheet 13.

The FPGA I/O supply U5 above remains 1.5 V.

CONFIGURATION IS MANUALLY HELD DURING INITIAL TESTS

Close J6 before applying power; it holds INIT_B low.

Check ALL rails at the FPGA before removing the shunt.

Regulator PG signals alone do not prove GTP voltage tolerance.

UNROUTED POWER PROPOSAL – NOT QUALIFIED

Sheet: /07 / I/O power; U4 now on sheet 13/
File: review_07_io_regulators.kicad_sch

Title: FPGA I/O supply; GTP supplies on sheet13

Size: A2 Date: 2026–09–29

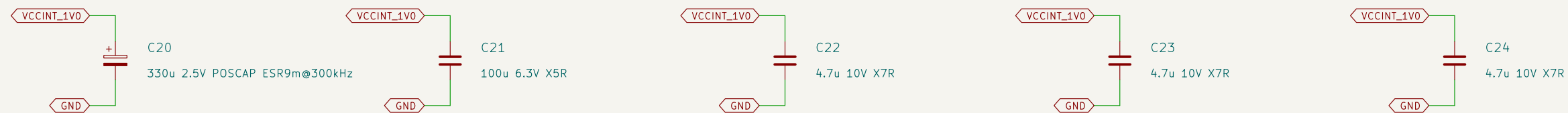
KiCad E.D.A. 10.0.6

Rev: R5 GTP power

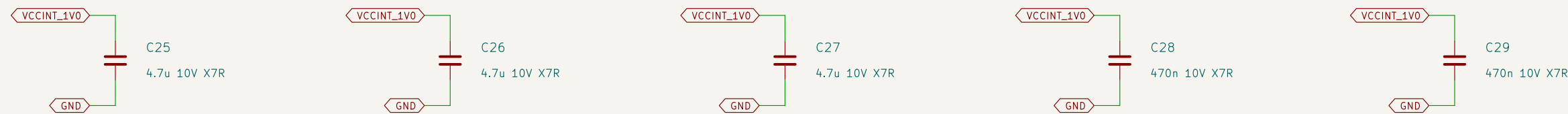
Id: 7/14

08 / Core decoupling – local charge for the 1.0 V rail

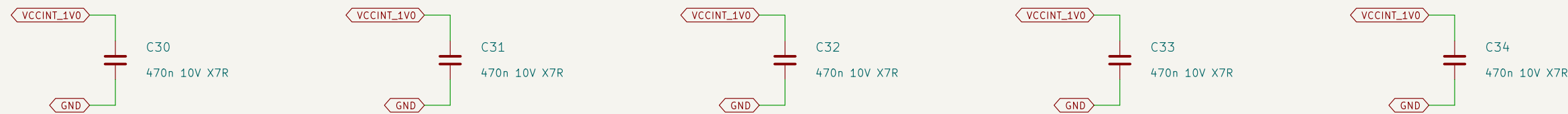
These capacitors sit between VCCINT_1V0 and ground. They support rapid FPGA current changes.



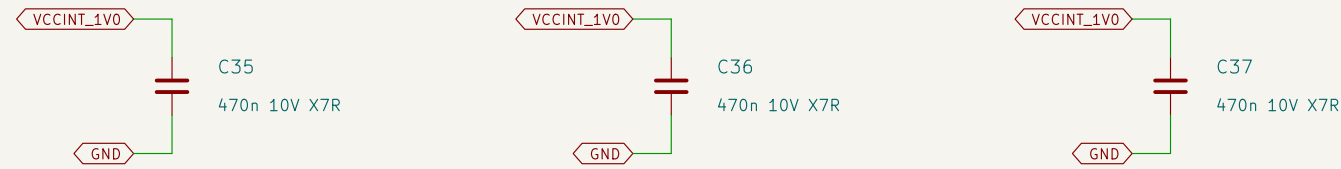
C20/C21 / BULK: larger charge reservoirs support the distributed rail.
C22–C27 / 4.7 uF: local bypass distributed near the FPGA power pins.



C28–C37 / 470 nF: small bypass capacitors support fast current changes.
The current loop through each capacitor and ground should be short.



All groups share the same rail; page spacing does not create separate supplies.
Capacitance value alone is insufficient: placement and rail impedance matter.



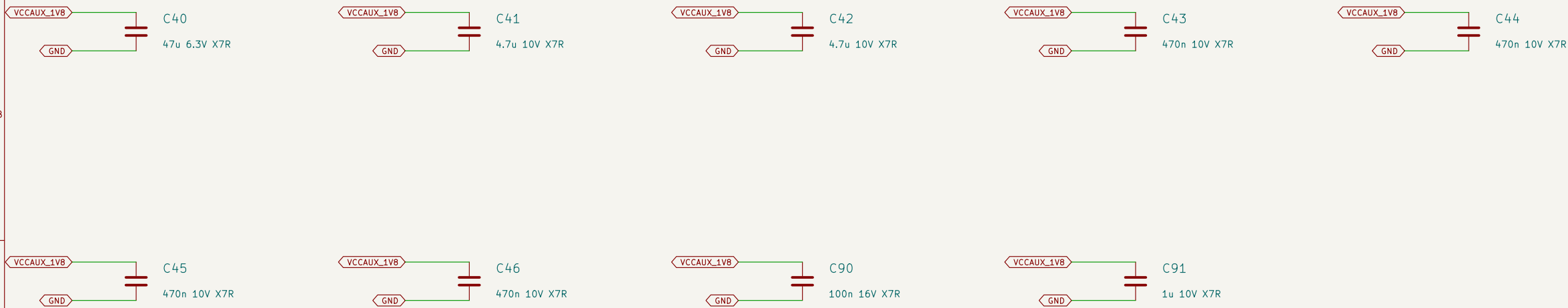
REVIEW: re–place the inherited capacitor groups for the new CSG325 power balls.
Confirm effective capacitance, regulator stability and voltage at the FPGA under load.

UNROUTED POWER PROPOSAL – NOT QUALIFIED		
Sheet: /08 / Core / BRAM decoupling review/ File: review_08_core_decoupling.kicad_sch		
Title: 08 / Core and BRAM decoupling – review placement next		
Size: A3	Date: 2026–09–29	Rev: R5 GTP power
KiCad E.D.A. 10.0.6		Id: 8/14

09 / Auxiliary and configuration bypass

Capacitor groups show placement intent; equal net names are one electrical supply across the board.

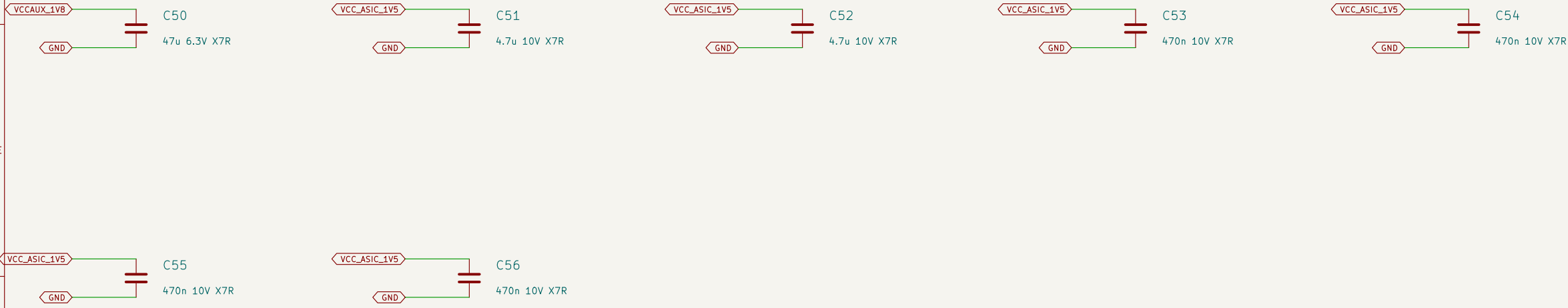
1.8 V / AUXILIARY GROUP



C40–C46 + C90/C91 / 1.8 V

These all connect to VCCAUX_1V8. Large capacitors hold local charge; smaller values support fast changes. Review placement near auxiliary/XADC power pins rather than treating this as an already isolated analog supply.

CONFIGURATION / SHARED 1.8 V GROUP

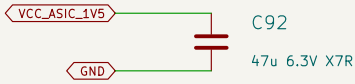


C50 / BANK 0 1.8 V; C51–C56 + C92 / BANK 14 1.5 V

These use the same auxiliary net. The configuration bank now uses 1.8 V. The three user I/O banks use 1.5 V. Redistribute inherited bank capacitors after reviewing the CSG325 pin map.

UNUSED 2.5 V GROUP REMOVED

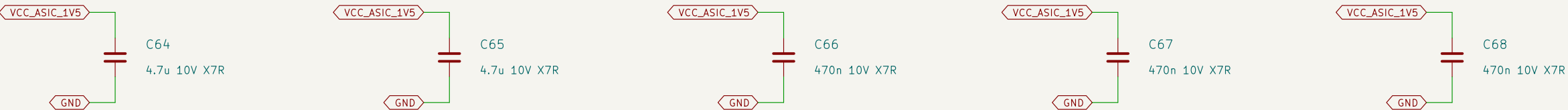
C57–C63 are no longer fitted. U4 now feeds GTP1.2 V on sheet13. New GTP bypass capacitors have separate analog supply nets.



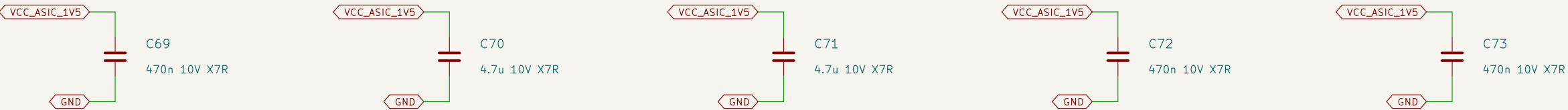
NEXT: assign each bypass group to its actual power balls and check the complete power–distribution network.

10 / ASIC-facing FPGA bank bypass – common 1.5 V rail

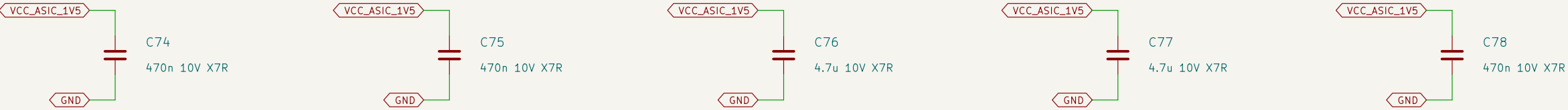
All C64–C82 use VCC_ASIC_1V5 and ground. This is the FPGA I/O supply, not power delivered to the ASICs.



4.7 uF GROUP / C64–C65, C70–C71 and C76–C77 provide local charge. Distribute these near the power pins of banks 14, 15 and 34.



470 nF GROUP / the smaller capacitors support rapid I/O switching currents. Short supply-to-capacitor-to-ground loops reduce voltage disturbance.



C82 / 47 uF bulk supports the shared rail; it does not isolate the banks. Rows are drawing groups. They are not separately regulated ASIC supplies.



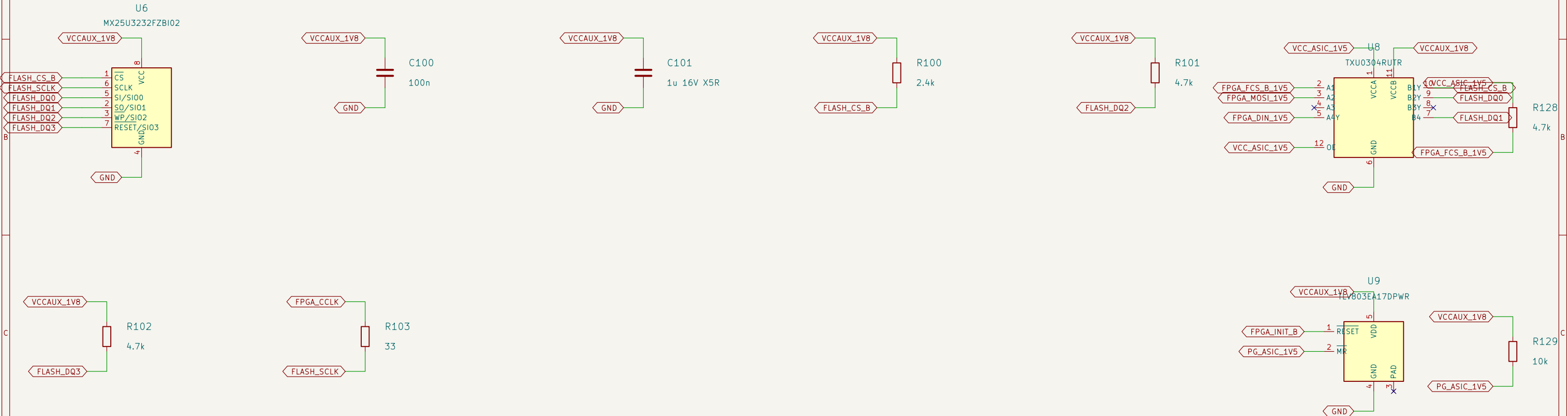
REVIEW: revise placement for the CSG325 package and the final carrier interface. Confirm simultaneous-switching load, return paths and voltage at the FPGA pins.

UNROUTED POWER PROPOSAL – NOT QUALIFIED		
Sheet: /10 / ASIC bank decoupling review/ File: review_10_asic_decoupling.kicad_sch		
Title: 10 / ASIC-facing FPGA bank decoupling – 1.5 V		
Size: A3	Date: 2026-09-29	Rev: R5 GTP power
KiCad E.D.A. 10.0.6		Id: 10/14

11 / 32 Mbit x1 SPI flash boot candidate – unrouted review

U6, U8 and U9 are fitted in this candidate; C101 stays DNP. Validate boot timing and rails before fabrication.

U6 / 1.8 V flash, 32 Mbit one image



FLASH + LOCAL SUPPORT

U6 holds one XC7A25T configuration image.
C100 bypasses flash; C101 is optional DNP.
R100–R102 pull CS/WP/RESET high at 1.8 V.
R103 drives flash SCLK from CCLK through 33 ohm.

U8 / BANK14 LEVEL SHIFT; U9 / STARTUP HOLD

U8 A=1.5 V, B=1.8 V: CS/MOSI to flash, SO to DIN.
U9 holds INIT_B until AUX and U5 PG settle.
M[2:0]=001 selects Master SPI x1; JTAG remains available.
J4 and the real power input are on sheet 6.

Boot timing, brownout recovery and package assembly need validation.

12 / GTP pins, filtered supplies and manual configuration hold

TX0/1/2 and RX1 lead to J4 through sheet 14. TX3 remains open; RX0/2/3 grounded.

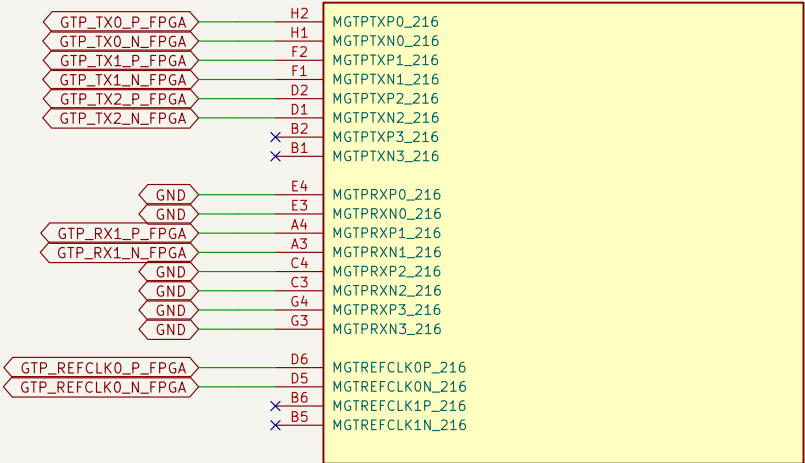
U1 / SERIAL SIGNAL PINS

TX0/1/2 and RX1 have a 100 nF AC-coupled custom J4 candidate on sheet 14.
MGTREFCLK0 receives the 125 MHz LVDS review clock on sheet 14.

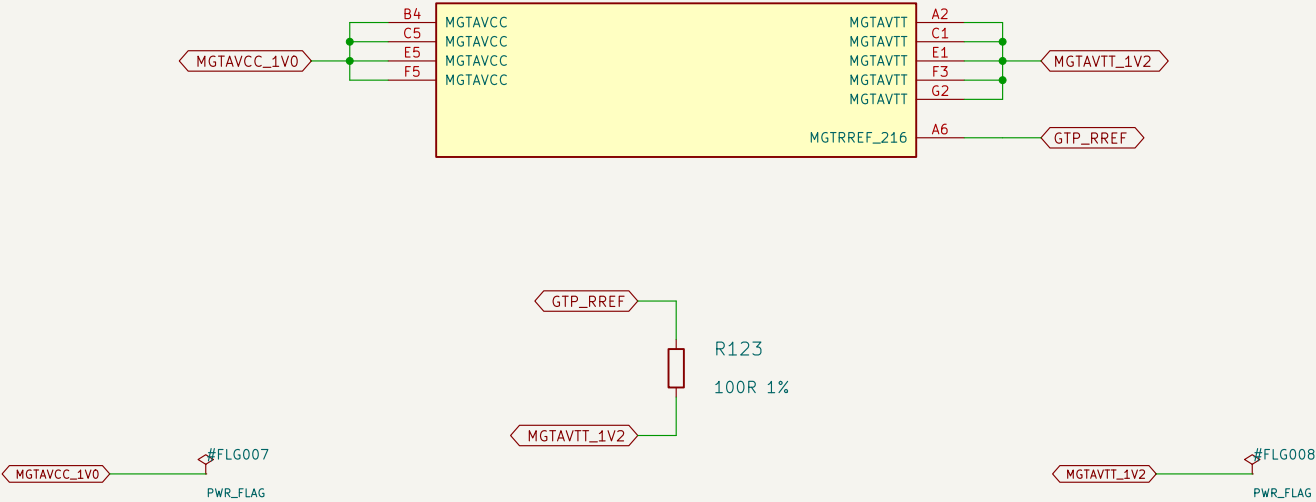
U1 / FILTERED ANALOG INPUTS

MGTAVCC_1V0 comes from U7 + FB1.
MGTAVTT_1V2 comes from U4 + FB2.

U1C
XC7A25T-2CSG325I



U1D
XC7A25T-2CSG325I



J6 / INSTALL SHUNT BEFORE POWER

INIT_B low delays configuration during startup.
Measure ALL rails at the FPGA in tolerance/stable.
Remove the shunt; only then begin JTAG configuration.
This is a manual prototype step, not a supervisor.

J6
INIT_B HOLD – shunt before power



UG482 Table 5-14: RX0/2/3 to GND; TX3 and REFCLK1 open. See coupling and Y2 on sheet 14.

R123 is 100 ohm / 1% from MGTREF to MGTAVTT. The sheet-14 GTP link is a review candidate; SI/jitter remain open.

UNROUTED POWER PROPOSAL – NOT QUALIFIED

Sheet: /12 / GTP pins + manual INIT_B hold/
File: review_12_gtp.kicad_sch

Title: GTP pins and manual configuration hold

Size: A3 Date: 2026-09-29
KiCad E.D.A. 10.0.6

Rev: R9a GTP contact map
Id: 12/14

13 / Dedicated GTP power – core supply remains separate

U7 is new. U4 is reused from the old 2.5 V circuit. These are populated circuit proposals, not measured power rails.

WHY TWO DEDICATED RAILS

AMD recommends no sharing
with non-transceiver loads.
U2 still powers core/BRAM.
The common INPUT can be shared.

STARTUP CONNECTIONS

- Core PG starts auxiliary U3.
- PG_AUX starts U7 and I/O U5.
- U7 PG then starts U4.
- New PG pull-ups use 1.8 V.
- Power-down order is not qualified.

AT EACH GTP SUPPLY GROUP

One 4.7 μF + two 100 nF caps.
Place at assigned supply balls.
FB1/FB2: 10 milliohm max DCR
at the specified test condition.
Verify hot/DC-biased performance.

VOLTAGE / NOISE ACCEPTANCE

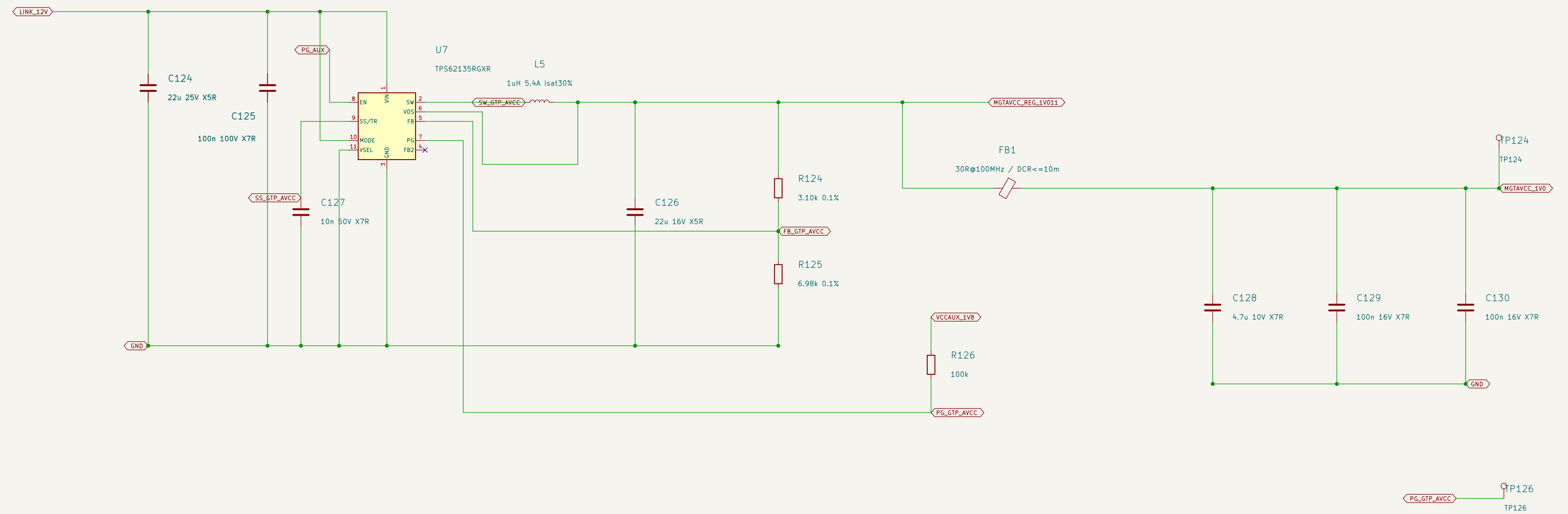
MGTAVCC: 0.97–1.03 V.
MGTAVTT: 1.17–1.23 V.
Noise target: below 10 mV p-p
at FPGA pins, 10 kHz–80 MHz.
Filtering is NOT yet measured.

J6 HOLD: SHEET 12

PG is not precision validation.
Measure all rails before JTAG.
XPE load and thermal budget TBD.

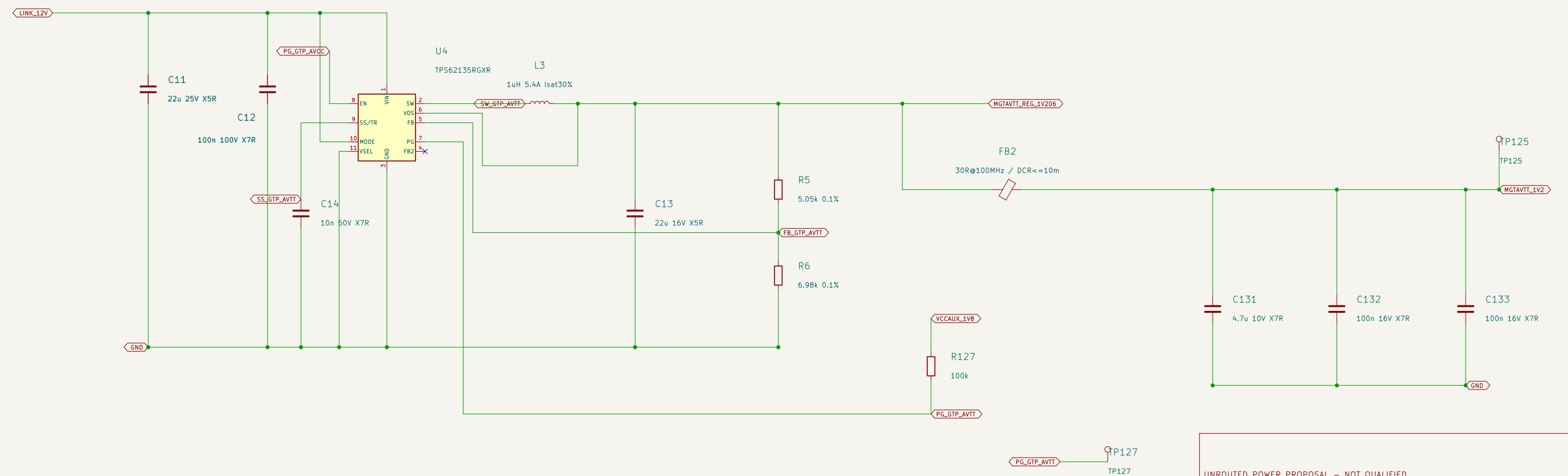
U7 / GTP ANALOG 1.0 V

Regulator setpoint: ~1.0109 V before FB1.
FB1 feeds C128 / C129 / C130 at the FPGA.



U4 / GTP TERMINATION 1.2 V

Regulator setpoint: ~1.2064 V before FB2.
FB2 feeds C131 / C132 / C133 and R123.



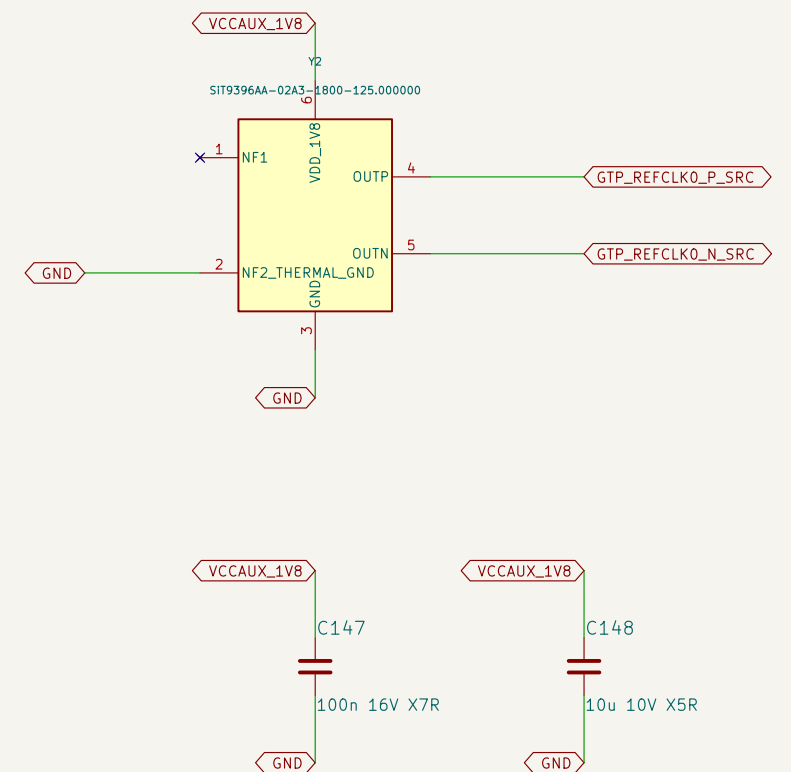
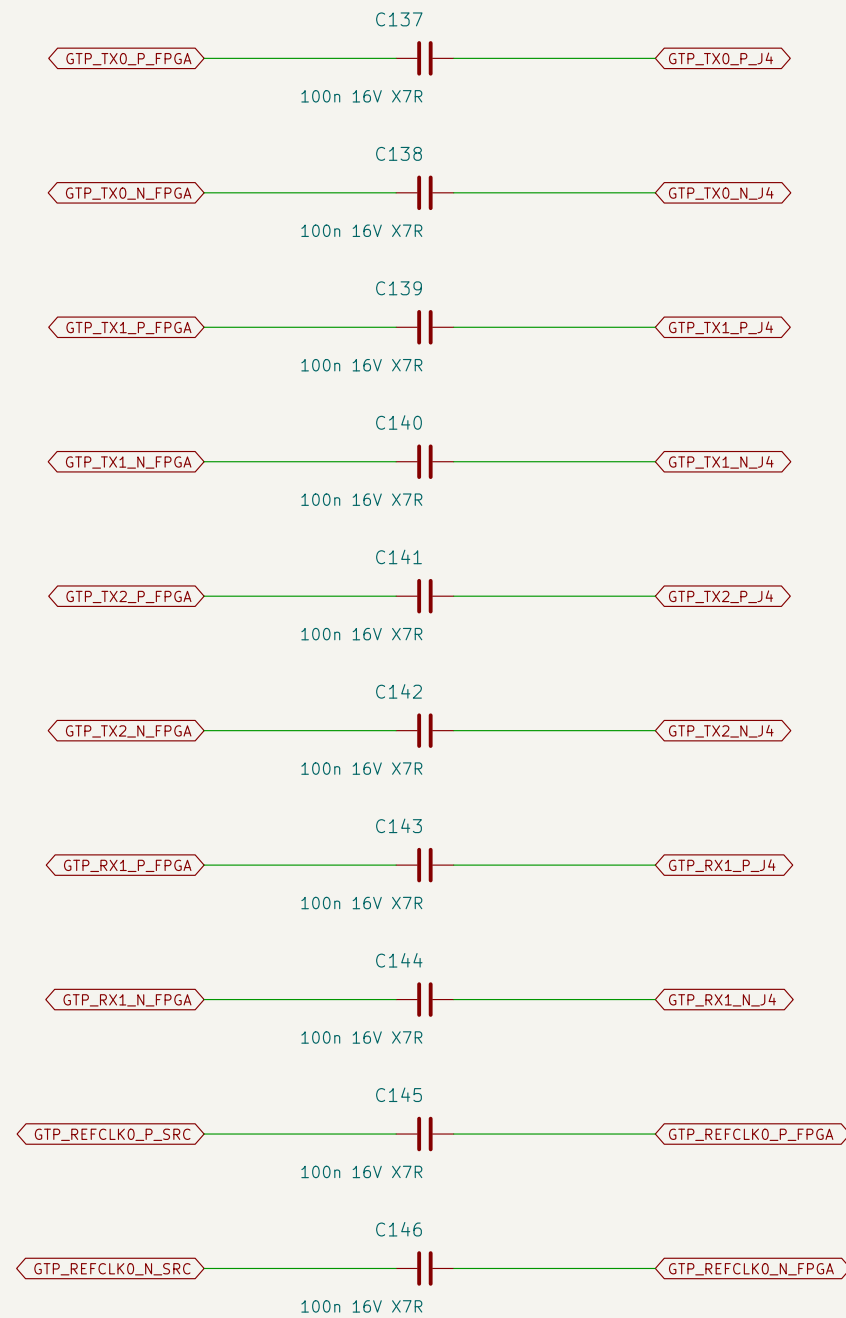
Nominal setpoints allow for filter drop. 1 A per rail is a calculation case only; current capacity is not approved.

UNROUTED POWER PROPOSAL – NOT QUALIFIED			H
Sheet: /13 / Dedicated GTP 1.0 V + 1.2 V power/ File: review_13_gtp_power.kicad_sch			
Title: 13 / Dedicated GTP 1.0 V and 1.2 V power – review proposal			
Size: A2	Date: 2026–09–29	Rev: R5 GTP power	
KiCad E.D.A. 10.0.6		Id: 13/14	

Ten 100 nF AC caps: TX0/1/2 x2, RX1 x2, LVDS REFCLK0 x2. Single coupling stage per conductor.

Connector TX uses contacts 3/5, 6/8, 9/11; control RX uses 12/14. Preserve JTAG and J4 pin19 12 V candidate.

Four-conductor cable channel, 100 ohm differential routing, receiver ownership and 125 MHz jitter need signoff.



Y2: exact SiTime order code; pin1 NF open, pin2 NF to GND, pin3 GND, 4 OUTP, 5 OUTN, 6 VDD.

C147 required local 100 nF. C148 follows SiTime best-jitter 10 uF note. Refclock-to-fabric remains Vivado TBD.

UNROUTED CANDIDATE; CABLE / SI / JITTER NOT QUALIFIED

Sheet: /14 / GTP AC coupling and 125 MHz reference/
File: review_14_gtp_link.kicad_sch

Title: 14 / GTP coupling and 125 MHz LVDS reference – review

Size: A3	Date: 2026-09-29	Rev: R9b GTP link
KiCad E.D.A. 10.0.6		Id: 14/14